

WEIMIN WU

Northwestern University • wwm@u.northwestern.edu • (+1) 773-681-4449

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EDUCATION

McCormick School of Engineering, Northwestern University, Evanston, USA Sep 2023 – Present
Ph.D. in Computer Science & Biochemistry and Molecular Genetics, Co-advised by [Han Liu](#) and [Yaping Liu](#)
McCormick School of Engineering, Northwestern University, Evanston, USA Sep 2023 – Jun 2026
M.S. in Computer Science, Advised by [Han Liu](#)
School of Mathematical Sciences, Fudan University, Shanghai, China Sep 2019 – Jun 2023
B.S. in Computational Mathematics, Outstanding Graduate of Fudan University, Advised by [Tao Chen](#)

RESEARCH INTERESTS

My research focuses on **AI for Medical Science**, integrating recursively self-improving AI agents with genomic foundation models to advance noninvasive early cancer detection using cell-free DNA and personalized cancer treatment through DNA/RNA vaccine sequence design. My existing work spans three areas: **(i) genomic foundation model development**, including GenomeOcean, a state-of-the-art pretrained genomic foundation model; **(ii) scientific applications**, including D-DOIT and DiMOS for DNA/RNA sequence design; and **(iii) genomic model serving infrastructure**, including Genome-Factory, an integrated platform for training, adapting, evaluating, and deploying genomic models.

GRANTS & AWARDS

NSF's NAIRR grant (NAIRR240377, \$460,000): Towards the world's leading and largest genomic foundation model.

2026 Ph.D. Student Research Award, Computer Science, Northwestern University

INTERSHIP EXPERIENCES

Genomics Deep Learning Engineering Intern | **NVIDIA**, CA, USA. Jun. 2026 – Sep. 2026
• Advisor: [Pankaj Vats](#), [Zhihan Zhou](#) & [Tong Zhu](#)
• Develop genomic foundation models for genome annotation.

Machine Learning Researcher | **Quest Diagnostics (Collaborate with NVIDIA)**, NJ, USA Jun. 2025 – Sep. 2025
• Advisor: [Paz Polak](#); Collaborator: [Pankaj Vats](#)
• Improve the existing method from 0.8 to 0.91 based on the genomic foundation model.

JOURNAL PUBLICATIONS

DNABERT-S: Learning Species-Aware DNA Embedding with Genome Foundation Models, Bioinformatics, 2025

Zhihan Zhou*, **Weimin Wu***, Zhong Wang, Han Liu et al. (*: joint first author)

A tailored and SOTA genomic model that develops species-aware embeddings for DNA sequences.

Uncovering the Genomic Manifold via Scalable Learning from the Global Microbiome, 2025

Weimin Wu*, Zhihan Zhou*, Yaping Liu, Han Liu, Zhong Wang et al. (Under Review of *Nature Methods*)

A 4B-parameter genomic generative foundation model that validates the Genomic Manifold Hypothesis.

CONFERENCE PUBLICATIONS

D-DOIT: Training-free Adaptation of Discrete Diffusion via Doob's h-Transform, NeurIPS, 2026

Jieke Wu*, Qijie Zhu*, **Weimin Wu***, Zeqi Ye, Minshuo Chen, Han Liu (*: joint first author)

Training-free guided discrete diffusion for regulatory DNA design and protein inverse folding.

Genome-Factory: An Integrated Library for Tuning, Deploying, and Interpreting Genomic Models, ICML, 2026

Weimin Wu, Zhong Wang, Han Liu et al.

The first integrated Python library for tuning, deploying, and interpreting genomic models.

Sci2Pol: Evaluating and Fine-tuning LLMs on Scientific-to-Policy Brief Generation, ICLR, 2026

Weimin Wu, Dashun Wang, Han Liu et al.

The first benchmark and training dataset to evaluate and fine-tune LLMs on scientific-to-policy brief generation.

In-Context Deep Learning via Transformer Models, ICML, 2025

Weimin Wu, Han Liu et al.

Prove the transformer's capability to simulate the training process of deep models via in-context learning.

In-Context Learning as Conditioned Associative Memory Retrieval, ICML, 2025

Weimin Wu, Han Liu et al.

The first interpretation of in-context learning from the perspective of dense associative memory models (Hopfield).

On Statistical Rates of Conditional Diffusion Transformers: Approximation and Estimation, ICLR, 2025

Jerry Hu*, **Weimin Wu***, Minshuo Chen, Han Liu et al.

The first comprehensive statistical theory (approximation and estimation) for conditional diffusion transformers.

On Statistical Rates and Provably Efficient Criteria of Latent Diffusion Transformers (DiTs), NeurIPS, 2024

Jerry Hu*, **Weimin Wu***, Han Liu

The first statistical (approximation and estimation) and computational theory for diffusion transformers.

Universal Approximation with Softmax Attention, ICML, 2026

Jerry Hu, **Weimin Wu**, Han Liu et al.

Prove that the attention layer is a universal approximator for sequence-to-sequence functions.

On Structured State-Space Duality, ICML, 2026

Jerry Hu, **Weimin Wu**, Han Liu et al.

Prove the equivalence between a structured state-space model and a masked attention mechanism.

Learning Manifold Data with Flow Matching, ICML, 2026

Sophia Pi, **Weimin Wu**, Han Liu et al.

A comprehensive statistical theory study of flow-matching transformers with low-dimensional manifold.

POLO: Preference-Guided Multi-Turn Reinforcement Learning for Lead Optimization, KDD, 2026

Ziqing Wang, **Weimin Wu**, Han Liu, Kaize Ding et al.

A sample efficient LLM framework for lead optimization by learning from full multi-turn optimization trajectories.

PREPRINTS

DiMOS: Doob-Guided Inference-Time Multi-Objective Search for Scientific Design, arXiv, 2026

Ziqing Wang*, Qijie Zhu*, **Weimin Wu***, Zeqi Ye, Minshuo Chen, Han Liu, Kaize Ding (*: joint first author)

A training-free method for multi-objective DNA, RNA, and protein design based on discrete diffusion.

Cell-JEPA: Latent Representation Learning for Single-Cell Transcriptomics, arXiv, 2026

Ali ElSheikh*, Rui-Xi Wang*, **Weimin Wu***, Aly A. Khan, Han Liu et al.

A joint-embedding predictive architecture for single-cell foundation model.

Discrete Flow Matching Policy Optimization, arXiv, 2026

Maojiang Su, **Weimin Wu**, Han Liu et al.

A unified reinforcement learning framework for fine-tuning discrete flow matching models.

Instance-aware Model Ensemble With Distillation For Unsupervised Domain Adaptation, arXiv, 2024

Weimin Wu, Tao Chen et al.

An adaptive ensemble method for unsupervised domain adaptation in computer vision.

INVITED TALKS

Feinberg School of Medicine, Northwestern University

• Advancing Science with World's Leading Genomic Foundation Models.

02/20/2026

School of Medicine, Shanghai Jiao Tong University

• Advancing Science with World's Leading Genomic Foundation Models

12/22/2025

International Society for Computational Biology, Oral

• DNABERT-S: Learning Species-aware DNA Embedding with Genome Foundation Models.

07/23/2025

Harvard Medical School, Harvard University

• DNABERT-S: Learning Species-aware DNA Embedding with Genome Foundation Models.

03/14/2024

ENTREPRENEURSHIP

Founder and President of Quantitative Investment Association (QIA), China

Jun. 2021 – Jun. 2023

• The largest non-profit organization in quantitative finance in China.

TEACHING EXPERIENCE

CS110: Introduction to Computer Programming, Northwestern University

Fall 2024

• Conducted weekly discussion sections and office hours for 100+ undergraduate students.

• Graded assignments and provided constructive feedback on Python programming projects.

PROFESSIONAL SERVICE

Reviewer of *ICLR 2026, NeurIPS 2025, ICML 2025, ICLR 2025*

PATENTS

Instance-aware Model Ensemble With Distillation For Unsupervised Domain Adaptation, 2023

Weimin Wu, Tao Chen et al., Chinese Patent, No. 202310743718.9

SUPERVISED RESEARCH ASSISTANTS

Maojiang Su (University of Science and Technology of China)

2025

• Next Position: CS Ph.D. at Northwestern University, First author paper @ ICML 2025

Morris Huang (National Taiwan University)

2024

• Next Position: CS Ph.D. at the University of North Carolina at Chapel Hill, First author paper @ ICLR 2025

Sophia Pi (Northwestern University)

2024

• Next Position: CS Ph.D. at University of Pennsylvania, One paper @ NeurIPS 2024